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2/26/2025

CS 6014

HW: Cryptography

1. Breaking a simple block cipher:

A block cipher with a block size of only 8 bits is very easy to break due to the small number of possible key combinations (2^8 , or 256). With a known plaintext it is easy to brute-force the key out for the known text because it is simple to try all 256 possible keys. Simply loop through all possible key combinations for a known ciphertext until the correct plaintext is decrypted, and then you have the key! This highlights the necessity of not using the same key twice, using large block sizes, and combining other strategies such as chaining to further mix up data that is being encrypted.

2. Block ciphers continued:

A: Since the same key is used, the same plaintext components will have the same ciphertext, allowing patterns in the data to be easily noticed. The eavesdropper can also potentially infer the block size since it is fixed.

B: If the attacker knows the structure of the data (i.e. block size), they could replace a block of ciphertext with their own version and it will not be noticed by the recipient as there are no checks in place. The attacker could also swap blocks in the ciphertext to mix up the message or target specific bits in block(s) of ciphertext to corrupt the message. Since there are no checks in place, the data will get through and be decrypted, but it will now be wrong, and Bob won't know.

C: Many approaches can prevent/counteract these attacks. Some approaches could include using a checksum or hash code of the message to ensure integrity and using block chaining when encrypting so individual blocks cannot be targeted.

3. Programming Part 1: A (bad) block cipher:

See code comments for answers to questions regarding the code.

4. Programming Part 2: RC4 Cipher:

See code comments for answers to questions regarding the code.